Krishna Tiwari

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EDUCATION

B.Tech, Computer Science & Engineering 2024 - 2028
CMR University

TRAININGS / CERTIFICATIONS

Theory Of Computation(TOC) / Automata : Complete Pack
Mar 2026 - Apr 2026

udemy, Virtual

Completed 12 total hours of comprehensive coursework covering automata and computational theory, instructed by Atchyut Kumar and Edufulness EFN

Introduction To The Fundamentals Of Databases
Feb 2026 - Mar 2026

SIMPLILEARN, Virtual

Successfully completed an online professional course focusing on the foundational concepts and skills required for database management

Fundamentals Of Operating Systems
Feb 2026 - Apr 2026

udemy, Virtual

Completed 22.5 total hours of instruction led by Hussein Nasser, covering the core principles and mechanics of operating systems

Software Engineering
Sep 2025 - Oct 2025

Infosys Springboard, Virtual

Successfully completed training focused on core software engineering methodologies, principles, and professional practices.

Python For Beginners
May 2025 - Jun 2025

SIMPLILEARN, Virtual

Successfully completed an online foundational course covering core Python programming concepts and beginner-level syntax to advance technical skills

The Complete Python Bootcamp From Zero To Hero In Python
Nov 2024 - Jan 2025

udemy, Virtual

Completed 22 total hours of comprehensive Python programming instruction led by instructors Jose Portilla and Pierian Training

PORTFOLIO

GitHub link ↗

PROJECTS

OrbitLab Repository & 3D Simulation | #Hackathon ↗
Dec 2025 - Mar 2026

- Technologies Used: JavaScript, HTML5/CSS3, 3D Simulation Libraries, Azure Static Web Apps, Git/GitHub.
- Collaborated on developing an advanced, interactive virtual satellite engineering platform designed to simulate orbital mechanics and satellite subsystems.
- Built and optimized integrated 3D simulation components to visually map orbital paths and hardware physics within a web-based interface.
- Configured the continuous integration and deployment (CI/CD) pipeline to host the platform architecture seamlessly on Azure Static Web Apps.
- Submitted the project to multiple premier technical competitions, managing repository branching and resolving cross-environment codebase bugs.

Expense Tracker Application | #3rd Sem project
Nov 2025

- Technologies Used: Java, Core Java Libraries, File I/O Streams.
- Developed a standalone desktop system engineered to monitor daily user financial logs, tracking income streams and localized expense data.
- Implemented robust file-handling mechanics (BufferedReader/BufferedWriter) to ensure data persistence across application restarts.
- Built a lightweight command-line/graphical interface focusing on clean string manipulation, error parsing, and algebraic financial aggregation.

FIRE AND GAS DETECTION SYSYTEM | #3rd Sem project|# Hardware

Oct 2025

- Technologies Used: Arduino Uno, C/C++, OpenCV, Python, Hardware Sensors, Serial Communication.
- Architected a real-time smart gas detection and automated warning system utilizing an Arduino Uno and physical gas sensors.
- Developed an interactive hand-tracking game by bridging Python-based computer vision (OpenCV) with microcontroller logic over serial communication interfaces.
- Debugged low-level firmware code, managed power distribution modules, and processed analog-to-digital sensor calibration data.

AI-Powered Nutrition Application | #Personal Project ↗

Apr 2026 - Present

- Technologies Used: React.js, Supabase, Google Gemini API, JavaScript, Tailwind CSS, Supabase Edge Functions.
- Engineered a full-stack, responsive web application using React.js and Tailwind CSS to allow users to conversationally log meals via natural language inputs.
- Built a secure backend architecture using Supabase Edge Functions (Deno) to interface with the Google Gemini API, protecting sensitive API keys from frontend exposure.
- Implemented structured JSON prompting to parse unstructured text inputs, automatically calculating precise ingredient weights, caloric data, and macronutrient breakdowns (proteins, carbs, fats).
- Designed relational database schemas in Supabase (PostgreSQL) to manage user authentication, session states, and historical dietary intake tracking.

Maze Runner Game | #2nd Sem project

Jun 2025

- Technologies Used: Python, Pygame, Object-Oriented Programming (OOP).
- Programmed a desktop-based game featuring dynamic maze generation algorithms, variable difficulty scaling, and a fully synchronized gameplay timer.
- Structured complex movement logic, automated enemy AI pathfinding, and state-driven win/lose event-handling mechanisms using Pygame.
- Refined user experience by implementing optimized graphics rendering, clean keyboard control layouts, and clear level-progression loops.

SKILLS

- JavaScript
- Java
- APIs
- Data Structures
- Agile Methodology
- SQL
- Python
- Microsoft Azure
- Database Management System (DBMS)
- Software Development Life Cycle (SDLC)
- C++ Programming
- Vibe Coding
- Generative AI Tools
- Canva
- Natural Language Processing (NLP)

EXTRA CURRICULAR ACTIVITIES

- ◦ Social Impact & Volunteering:
  Volunteered on the ground for 4 days (21st to 24th December 2025) in the Pulse Polio immunization campaign Karnataka 2025, supporting community health administration.

  ◦ Community Advocacy:
  Participated in the Walk to EmpowerHER walkathon organized by Child Rights and You (CRY) to support and promote girls in STEM education.
- ◦ College Competitions:
  Secured 1st Place in the college Treasure Hunt competition and achieved 2nd Place in the college Cricket tournament.

  ◦ Technical Seminars & Workshops:
  Actively attended multiple academic seminars covering advanced industry domains, including Cyber Security, Ethical Hacking, Startup Building, and the Future of Generative AI Tools.

ADDITIONAL DETAILS

- HACKATHON |India Innovation 2026|#OrbitLab Led a 4-member team across the full SDLC. Directed system architecture, developed the 3D web frontend, structured backend logic, and managed Azure deployment, driving the project from ideation to submission

- HACKATHON |AMD Slingshot Hackathon|#OrbitLab Spearheaded a team of 4 through all development stages. Architected the core platform, built interactive UI/UX, managed Git version control, and executed the final cloud deployment under strict deadlines.
- HACKATHON |Microsoft Imagine Cup |#OrbitLab Directed a 4-person team across the entire project lifecycle. Built the cloud architecture, engineered the frontend interface, integrated AI into the backend, and handled all final hosting and deployments.